QP CODE : 25900060

MGU-UGP (HONOURS)REGULAR EXAMINATION, MARCH2025
SECOND SEMESTER
MG2DSCECT100, MG2DSCECC100 - ESSENTIAL CONCEPTS IN DIGITAL ELECTRONICS
(2024 Admission Onwards)
Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Interest(I), Appreciation(Ap) and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability

SCHEME OF EVALUATION

PART A
Multiple Choice Questions
Answer all questions. Each carries 1 mark

		Learning Domain	CO No(s)
1	a. 10010	U	[CO1]
2	c. Universal Functionality	K	[CO2]
3	b. (11010100)2	U	[CO1]
4	c. Digital Watch	K	[CO1]
5	a. 1	U	[CO2]
6	a. 2	K	[CO1]
7	c. 10110	U	[CO1]
8	a. AND	U	[CO2]
9	a. Adds two binary digits	K	[CO3]
10	c. The SR latch has two inputs and the SR flipflop has three inputs including clock	K	[CO3]

PART B
Short Answer Questions
Answer 4 questions. Each carries 5 mark

11	a) 59 b)55 For each answer 2.5 mark	U	[CO1]
12	Any attempt give 5 mark (Question wrong)	U	[CO2]
13	Any 5 application give 5 mark	K	[CO3]
14	NOR gate - Symbol, Expression and Working -3 Mark Truth Table- 2 Mark	K	[CO2]
15	DeMultiplexer -Tuth Table -2Mark Diagram and Working - 3 Mark	K	[CO3]
16	a)57=1010111 b)3A5=1110100101 c)F80B=1111100000001011 For correct answer = 5 Mark (1.5+1.5+2)	K	[CO1]

PART C

Essay Questions

Answer any 2 questions. Each carries 10 mark

17	For each gate 2.5 marl respectively (2.5 x 4 =10) symbol - 0.5 mark Truth Table - 1 Mark Logic Expression - 1 Mark	U	[CO2]
18	Defenition - 2 mark Diagram - 4 Mark Working - 4 mark	K	[CO3]
19	Two forms of DeMorgan's Law - 5 Mark each (2 x 5 =10) Statement- 2 mark Expression- 1 Mark Example - 2 mark	U	[CO2]
20	i)7A5B =31323 and 82BC = 33468 (2 x1 =2 marks) ii)3143= 1635 and 3475 = 1853 (2 x1 =2 marks) iii) 4096 = 1000 and 5678 = 162E (2 x1 =2 marks) iv) 111000 = 70 2 mark v) 4CDA=19674 and 89FB=35323 (2 x1 =2 marks)	K	[CO1]